Team Charter

Team Members: Xiyu Wang, Ivy Wang, Emily Chen, Tracy Che

Team Name: WangWangCheChen

1. Team Purpose or Mission

We hope to build a web-page embedded with a story line to show our data visualization and analysis results.

The expected output is going to be a public-phased webpage that allows both public audiences who are interested in solar implementation in America and professionals from both academia and industry who would like to explore the trends of the market able to learn about Solar Market in America (North America and South America), especially the role of China's export solar. Straightforward visualizations will be helpful to read and understand. The audience will expect the output in clear and professional analytical quality.

Our combined skills in R and Python would give the group flexibility in choosing between languages of data analysis. We also have experience with various data visualization and deliverables, such as StoryMap and webpage construction.

Some of our group members are more professional in the energy market, bringing deeper insight and context about what to research. The other group members are actively exploring the topics so they can also bring innovations that combine with other fields such as economics and ecosystems.

Research Scope:

- Study Area: North America (USA, Canada, Mexico), South America
- Trade Focus: Solar photovoltaic (PV) exports from China to countries in the study area only
- Primary Datasets Used: Ember China Solar PV Export dataset and IRENA installed capacity data
- Field of Study: China's solar PV exports and installed solar PV capacity in the study countries

Main Research Question:

- How have China's solar PV exports to North and South American countries evolved over time?
- How do these trends relate to changes in installed solar PV capacity in these countries?
- How

Supporting Questions:

- How do differences in solar PV import tariffs and trade policies across countries in the study area relate to Chinese solar PV export trends?

- How have changes in solar PV import tariffs and trade restrictions coincided with changes in installed solar PV capacity in selected countries?

How might differences in national renewable energy policies and incentives help explain variations in solar PV capacity growth across the study area?

2. Communication Plan and Collaborative Coding

Meeting and Communication:

- We will hold weekly meetings every Thursday during class time to discuss project progress, assign tasks, and resolve any issues.
- Additional meetings will be scheduled outside of class hours based on team members' availability if needed.
- We primarily communicate through a WeChat group and by email, and team members are expected to respond within 24 hours. In case of urgent matters, please contact us via WeChat as soon as possible.
- After each meeting, we will document key decisions, task assignments, and next steps in a shared document or task list to ensure all members are informed.

Collaborative Coding and File Sharing:

- Our team will use Python and R for data cleaning, analysis, and visualization.
- We will use a shared GitHub repository to store and manage code, datasets, and project documentation.
- Team members are required to regularly upload their work to GitHub and communicate major changes to avoid code conflicts and ensure the reproducibility of our research.

Individual team members' code should be annotated with comments as needed to ensure it is understandable to everyone.

3. Roles and Responsibilities

- Project Interest & Skills: Xiyu has experience using Python and R and is familiar with environmental data analysis. These skills will help the team with data exploration and visualization. Through this project, she hopes to gain a deeper understanding of the international solar energy market and the relationship between trade policies and renewable energy deployment.
- Tracy has experience using Python and R Studio. She has experience analyzing data and providing visualizations. She is new to solar power systems but is enthusiastic to explore energy data analysis. With her sufficient environmental and ecological background with skills with data processing, she can contribute to the methodologies in the group project.
- Emily has experience in Python and R Studio. She will utilize her skills in data analysis cross-check with teammates' results in Python. She also has experience in webpage creation so that she will create the Google site to produce the final output deliverable.
- Ivy has experience using R Studio for environmental data analysis, statistical analysis, and visualization. She has also developed Python skills through ENVIRON 700. In

addition, her experience using Excel for energy and carbon market analysis could help with organizing export data.

- Individual Responsibilities:
    - Ivy: Data cleaning and analysis
    - Tracy: Communications with instructor, any helps with R visualizations
    - Emily: Use R to create plots and create the google sites
    - Xiyu: Data visualization and Data analysis and visualization using visualization using Python and R

- Shared Responsibilities:
    - All team members will contribute to refining research questions, reviewing data quality, discussing analytical findings, and reviewing the final project deliverables. Members will also collaborate to ensure the accuracy, clarity, and consistency of the analysis and visualizations.

4. Operating Norms or Ground Rules

Before our weekly group meeting, everyone is expected to complete their assigned tasks. After modifying the code, each person should provide well-organized files, readable code (including comments to help others understand it), and documentation detailing the decisions made during data cleaning. If anyone is unable to attend a meeting, they should notify the other team members in advance.

Rotating submission based on anyone who is responsible for the last steps. Always ensure including all the names and communicating before submission. Everyone must reply before submitting the assignment (BEFORE DDL).

5. Values and Behaviors

We value reliability, respectful communication, and openness to different ideas. Everyone should complete their assigned tasks on time, communicate their progress, and reach out to teammates any time when needed or in emergencies.
We encourage everyone to share their opinions and ask questions during meetings or in our WeChat group. If someone is having difficulties, we should figure the issue out together and offer support.
If there are concerns about uneven participation of the quality of work, we will first communicate openly within the group, and solve the problems by identifying the space of improvement.

6. Resources and Constraints

Main resources:

1. Ember China Solar PV Export Dataset: Provides data on China's solar PV exports, including destination countries and export-related information.
2. IRENA Installed Capacity Dataset:Provides annual installed solar PV capacity data for countries in our study area.

Other resources include course materials, data documentation, guidance from data providers and course instructors, and any future supplemental datasets to fit our data visualization.

Main Constraints:

- Time Limitations: The six-week project timeline might limit the depth of our analysis and the complexity of the visualizations we could develop. We will prioritize our main research questions and essential visualizations to ensure that the project can be completed on time.
- Data Availability and Compatibility: The Ember and IRENA datasets might differ in their temporal and geographic coverage, which could limit comparisons across countries. We need to identify overlapping countries and years to ensure meaningful analysis.
- Geographic Scope: Our study area includes North and South America, but analyzing every country might not be feasible within the available time. We may narrow our focus to selected countries based on data availability and relevance to our research questions.

Unanswered Questions and Limitations:

At the beginning stage, our team needs to determine whether the datasets provide sufficient overlapping data to compare Chinese solar PV exports with installed solar PV capacity across our study area. Our team also recognizes that changes in solar PV exports might not directly correspond to changes in installed capacity within the same year.

If our team includes the trade policies or tariffs in our analysis later, we will need additional data sources to understand policy differences across countries.

To address these uncertainties, our team will review the dataset documentation and seek guidance from the instructor or teaching assistant if necessary. We will document important assumptions and data limitations during our analysis and communicate them clearly in our final webpage and visualizations.

7. Decision-Making Guidelines

We will aim to reach consensus through group discussions. Major decisions regarding research questions, datasets, analytical methods, and dashboard design will be discussed during our weekly meetings, while minor decisions can be made through our WeChat group. We will consider relevant data, available evidence, and all team members' perspectives when making decisions.

If we cannot reach a consensus, we will make decisions through a majority vote after considering everyone's opinions.

All team members should have the opportunity to participate in major decisions. However, if a member does not respond within 24 hours or an approaching deadline requires a timely decision, the remaining members may proceed after making reasonable efforts to communicate with everyone.

We will consult the instructors when we encounter major data issues such as merging problems or when we are uncertain about research frames.

8. Goals and Objectives

Our goal is to develop an accessible and informative webpage that showcases trends in China's exports of photovoltaic (PV) products to countries in North and South America, as well as the status of PV installed capacity in those countries. This webpage will help users explore differences among countries and understand the potential links between PV trade, installed capacity, and related energy policies.

Main Objectives:

1. Data Cleaning and Standardization: Ensure that the dataset has been properly cleaned and organized, and that it remains consistent across different countries and years.
2. Meaningful Data Analysis: Conduct analyses addressing our core and supporting research questions, including comparisons of PV exports and installed capacity across countries within the study region.
3. Clear and Comprehensive Data Visualization: Create accurate and easy-to-understand visualizations that effectively present key trends and research findings.
4. Easily Accessible Deliverables: Create well-structured, user-friendly web pages that allow the target audience to easily explore and understand the research findings.
5. Accuracy of Data and Results: Verify data processing workflows, calculation methods, and visualizations to ensure the reliability of the research conclusions.

Evaluation Criteria:

We will verify the completeness and consistency of the data, review the accuracy of the analysis and visualization results, and ensure that the research findings address the established research questions.

In addition, we will test the website's functionality, navigation design, and accessibility to ensure that users can easily explore and understand the information presented.

Deliverables:
Our core deliverables include cleaned datasets, reproducible analysis scripts, and a webpage that presents the key research findings through clear visualizations. If time permits, we might also develop additional interactive features, such as a “Story Map.”

9. Schedule and Milestones

Week	Main Tasks	Lead	Reviewer
Week 1: Project Setup and Research Scope	Identify the research question and research area, assess data feasibility, finalize the team charter, set up a shared GitHub repository, document the dataset information, and complete the initial data import.	Sabrina	Tracy
Week 2: Data Preparation	Inspect, clean, and organize the Ember and IRENA datasets. Identify missing values and verify data consistency.	Ivy	Sabrina
Week 3: Exploratory Data Analysis	Examine trends in China's photovoltaic exports and installed capacity, comparing data across different countries and time periods.	Emily	Ivy
Week 4: Visualization and Webpage Development	Use Python and R to create preliminary visualizations, identify key findings, and build a web prototype using Google Sites.	Tracy	Emily
Week 5: Validation and Refinement	Verify the accuracy of the data, review the analysis results, and test the website's functionality. Incorporate all of the above feedback and refine the visualization and analysis.	Sabrina and Ivy	Tracy and Emily

Week 6: Final Presentation and Submission	Complete the final revisions, verify reproducibility, and check GitHub access permissions. Prepare and rehearse the presentation, then submit the final project deliverables.	Tracy and Emily	Sabrina and Ivy

10. AI disclosure

We used ChatGPT to assist with brainstorming initial research scopes and troubleshooting code. All analytical decisions, results, and interpretations were made and reviewed by our team manually.